

A character extends to a pure state

`MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_extension`

theorem

Chooses an extreme state in the compact face of all extensions of one prescribed character.

Statement

Let A be a nonzero unital complex C^* -algebra. Let $D \subseteq A$ be a norm-closed unital $*$ -subalgebra and $\chi : D \rightarrow \mathbb{C}$ a character. There exists a continuous complex-linear functional φ on A such that it is a state, is pure, and satisfies

$$\varphi(d) = \chi(d) \quad (d \in D).$$

A state is positive and normalized at the unit; purity means extremality in the real convex state space.

Assumptions

No commutativity assumption is made on D . Its closedness and the specified character are inputs. Neither algebra separability nor a Hilbert-space representation is assumed.

Conclusion

One pure state of A extends that same character on every element of D .

Proof route

Use Hahn–Banach to obtain a state extension, show that all extensions form a nonempty compact face, and choose a pure state in that face.

▼ Proof steps

1. Write $S(A)$ for the state space of A . The character χ is a positive normalized functional of norm one on D . Apply the norm-preserving Hahn–Banach extension in A norm-preserving Hahn–Banach extension is an ambient state to the underlying complex subspace $D \subseteq A$ and this functional. Obtain $\psi : A \rightarrow \mathbb{C}$ with $\psi|_D = \chi$ and $\|\psi\| = 1$. Since $1_A \in D$, $\psi(1_A) = \chi(1_A) = 1$. The C^* -algebra criterion $\|\psi\| \leq 1$ and $\psi(1) = 1$ implies positivity. Thus the set

$$F_\chi = \{\psi \in S(A) : \psi|_D = \chi\}$$

of state extensions is nonempty.

2. Give the continuous dual the weak-star topology. Each condition $\psi(d) = \chi(d)$ is closed because evaluation at a fixed d is continuous, and the state conditions are weak- $*$ -closed. States have norm at most one, so the closed set F_χ lies in the weak-star compact dual unit ball and is compact. It is convex because

positivity, normalization and all restriction equations are preserved by real convex combinations. These are the compactness and nonemptiness facts in Compactness, nonemptiness and the face property.

3. The character is pure on D by A character is a pure state. Explicitly, if $\chi = (1 - t)\alpha + t\beta$ with states on D and $0 < t < 1$, for an arbitrary $d \in D$ put $q = d - \chi(d)1$. Then $\chi(q^*q) = 0$, while both $\alpha(q^*q)$ and $\beta(q^*q)$ are nonnegative. The convex equality forces both to vanish. Cauchy–Schwarz gives $|\alpha(q)|^2 \leq \alpha(q^*q)\alpha(1) = 0$ and the corresponding inequality for β . Thus $\alpha(q) = \beta(q) = 0$, so normalization gives $\alpha(d) = \beta(d) = \chi(d)$ for every d .

4. Let $\psi_0, \psi_1 \in S(A)$ and $0 < t < 1$ satisfy

$$(1 - t)\psi_0 + t\psi_1 \in F_\chi.$$

Their restrictions to D are states, and restriction of this equation gives

$$(1 - t)\psi_0|_D + t\psi_1|_D = \chi.$$

Step 3 implies $\psi_0|_D = \psi_1|_D = \chi$, so $\psi_0, \psi_1 \in F_\chi$. This is exactly the face property proved in Compactness, nonemptiness and the face property.

5. The extreme-point existence theorem for a nonempty compact convex set in the real locally convex weak-star dual gives a state $\varphi \in \text{ext}_{\mathbb{R}} F_\chi$. Since F_χ is a face of $S(A)$, the same φ is extreme in $S(A)$. Its membership in F_χ gives positivity, $\varphi(1) = 1$ and $\varphi(d) = \chi(d)$ for every $d \in D$; its extremality gives purity. These are all the required conclusions for one continuous complex-linear functional.

Main citations

- Exact theorem and proof
- A norm-preserving Hahn–Banach extension is an ambient state
- Compactness, nonemptiness and the face property
- A character is a pure state
- The pure-state predicate

Lean source signature (exact)

```
theorem exists_pureState_extension (D : StarSubalgebra ℂ A) [IsClosed (D : Set A)]
  (chi : WeakDual.characterSpace ℂ D) :
  ∃ phi : A →L[ℂ] ℂ,
    phi ∈ stateSpace A ∧ IsPureState A phi ∧ ∀ d : D, phi d = chi d
```

In the source	Mathematical meaning
$(D : \text{StarSubalgebra } \mathbb{C} A)$	A unital complex $*$ -subalgebra D of the nonzero unital ordered C^* -algebra A .
$[\text{IsClosed } (D : \text{Set } A)]$	The carrier of D is norm closed in A ; no commutativity binder is present.

In the source	Mathematical meaning
<code>(chi : WeakDual.characterSpace C D)</code>	The prescribed nonzero continuous complex character χ of D .
<code>∃ phi : A →L[C] C</code>	Choose one continuous complex-linear functional φ on all of A .
<code>phi ∈ stateSpace A</code>	It is positive on positive elements and satisfies $\varphi(1) = 1$.
<code>IsPureState A phi</code>	The same state is extreme for real convex combinations in $S(A)$.
<code>∀ d : D, phi d = chi d</code>	For every element d of the given subalgebra, evaluating the same φ at its ambient element equals the original character value $\chi(d)$.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_extension

Accepted content SHA-256: c0c91fda02ed2ed8ec467f2cb43fc89e00623ec94115b0726e913c4fee64f040

Accepted source guide SHA-256: 3ff75ee29922635da168954add276b03e81e8cca1a487a26a731f7a7933b2c48

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73